

newdb Shared Bundle External Integration Manual (Pure LaTeX)

newdb Project

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1 Purpose and Scope

This document is a pure LaTeX technical manual for integrating `newdb` shared libraries from external language runtimes. It covers:

- Build and distribution model for `shared_bundle`
- Full C API and gtest C API references
- Command-driven execution model and templates
- Unified result parsing and error handling
- Deployment and operational recommendations

2 Shared Bundle Model

2.1 Output directories

- VS/MSVC: `build/newdb-vs2026-mt-tests/shared_bundle/Release`
- MinGW: `build/newdb-mingw/shared_bundle/Release`
- WSL/Linux: `build/newdb-wsl-tests/shared_bundle/Release`

2.2 Build commands

```
# VS/MSVC
cmake --build build/newdb-vs2026-mt-tests --config Release --target shared_bundle --parallel

# MinGW
cmake --build build/newdb-mingw --target shared_bundle --parallel

# WSL/Linux
cmake --build build/newdb-wsl-tests --target shared_bundle -j8
```

2.3 Dependency precedence policy

- Windows bundles include `run_*.cmd` launchers to prioritize local DLLs.
- Linux bundles are linked with `RPATH=$ORIGIN` to prefer colocated `.so`.
- MinGW bundle includes runtime DLLs:
 - `libstdc++-6.dll`
 - `libgcc_s_*.dll`
 - `libwinpthread-1.dll`

3 newdb C API Reference

3.1 Header and compatibility constants

Header path:

- `newdb/engine/include/newdb/c_api.h`

Version/ABI constants:

- NEWDB_C_API_VERSION_MAJOR/MINOR/PATCH
- NEWDB_C_API_ABI_VERSION

3.2 Error codes

Code	Description
NEWDB_OK	Success
NEWDB_ERR_INVALID_ARGUMENT	Invalid argument or buffer contract violation
NEWDB_ERR_INVALID_HANDLE	Invalid or destroyed session handle
NEWDB_ERR_EXECUTION_FAILED	Command/business execution failed
NEWDB_ERR_INTERNAL	Internal engine failure
NEWDB_ERR_LOG_IO	Log or file I/O failure
NEWDB_ERR_SESSION_TERMINATED	Session terminated and cannot continue

3.3 Thread safety contract

- Global metadata APIs are thread-safe.
- Per-session APIs are safe only when one thread owns one handle.
- Concurrent use of the same handle is not guaranteed safe.

3.4 Exported functions

Global/metadata

- newdb_version_string
- newdb_api_version_major, newdb_api_version_minor, newdb_api_version_patch
- newdb_abi_version
- newdb_negotiate_abi
- newdb_error_code_string
- newdb_sum

Schema/session/execution

- newdb_check_schema_file
- newdb_session_create
- newdb_session_destroy
- newdb_session_set_table
- newdb_session_last_error
- newdb_session_execute
- newdb_session_runtime_stats
- newdb_session_append_runtime_snapshot

3.5 Recommended lifecycle

1. Load library by absolute path.
2. Verify ABI via `newdb_negotiate_abi`.
3. Create session.
4. Execute command stream.
5. Capture runtime stats/snapshots if needed.
6. Destroy session.

4 `gtest_capi` Reference

Header path:

- `newdb/include/gtest_capi.h`

Main API groups:

- Init/Run: `gtest_capi_init_from_argv`, `gtest_capi_run_all`
- Discovery: `gtest_capi_list_tests`, `gtest_capi_enumerate_tests`
- Options: filter/output/repeat/shuffle/random_seed/color and failure/exception controls
- Generic option bridge: `gtest_capi_set_option`, `gtest_capi_get_option`
- Counters: suite/test totals, pass/fail/skip/disabled and elapsed time

5 Feature-Oriented Integration Guidance

Through `newdb_session_execute`, external systems can drive:

- DDL/schema management
- DML and query operations
- Transaction and conflict/isolation controls
- WAL/recovery controls
- Runtime maintenance controls (vacuum, group commit, hot index)

Recommended wrapper layers:

- DDL wrapper
- DML wrapper
- Query wrapper
- Txn wrapper
- Runtime tuning wrapper

6 Command Templates for newdb_session_execute

6.1 DDL and schema

```
CREATE TABLE(users)
USE(users)
SET PRIMARY KEY(uid)
CREATE SCHEMA(finance_v1)
DROP SCHEMA(finance_v1)
ALTER TABLE users SET SCHEMA(finance_v1)
ALTER TABLE users REMOVE SCHEMA
RENAME TABLE(users_v2)
DROP TABLE(users_v2)
```

6.2 DML and query

```
DEFATTR(name:string, balance:int, dept:string)
INSERT(1, alice, 1000, rd)
UPDATE(1, alice, 1200)
DELETE(1)
DELETEPK(alice)
SETATTR(1, dept, infra)
DELATTR(dept)
RENATTR(balance, amount)
FIND(1)
WHERE(balance, >, 1000, AND, dept, =, rd)
COUNT
PAGE(1, 20, id, desc)
SUM(balance)
AVG(balance)
MIN(balance)
MAX(balance)
```

6.3 Transaction and runtime tuning

```
BEGIN
SAVEPOINT(sp1)
ROLLBACK TO(sp1)
RELEASE SAVEPOINT(sp1)
COMMIT
TXNISOLATION snapshot
TXNISOLATION read_committed
WRITECONFLICT reject
WRITECONFLICT wait 2000
AUTOVACUUM threshold 500
AUTOVACUUM interval 60
WALSYNC normal 200
WALADAPTIVE on
GROUPCOMMIT 10 128
HOTINDEX on
```

7 Unified Result Parsing Standard

Always evaluate:

1. Return code (`rc`) from `newdb_session_execute`
2. `newdb_session_last_error`
3. Human-readable `output_buf`

Recommended result DTO:

```
ExecuteResult {  
  rc: int  
  last_error: int  
  error_name: string  
  output: string  
  ok: bool  
}
```

Buffer sizing guideline:

- 4KB minimum for simple commands
- 16KB+ for complex query output

8 Deployment and Operations

8.1 Deployment checklist

1. Distribute full `shared_bundle/Release` directory.
2. Use absolute-path dynamic loading.
3. Verify ABI before first session creation.
4. One thread per session handle.
5. Record `rc` + `last_error` + `output` for all commands.

8.2 Observability

Collect:

- Commit latency and conflict metrics
- WAL recovery elapsed/scanned records
- Vacuum trigger count and reclaimed bytes

9 Troubleshooting

- “Module not found”: missing colocated dependencies in bundle.
- Wrong DLL/SO loaded: enforce absolute path and launcher usage.
- WSL googletest download issues: use local `FETCHCONTENT_SOURCE_DIR_GOOGLETEST`.